

Assignment module 6: Network Security, Maintenance, and Troubleshooting Procedures

Section 1: Multiple Choice

1. What is the primary purpose of a firewall in a network security infrastructure?

- a) Encrypting network traffic
- b) Filtering and controlling network traffic**
- c) Assigning IP addresses to devices
- d) Authenticating users for network access

2. What type of attack involves flooding a network with excessive traffic to disrupt normal operation?

- a) Denial of Service (DoS)**
- b) Phishing
- c) Spoofing
- d) Man-in-the-Middle (MitM)

3. Which encryption protocol is commonly used to secure wireless network communications?

- a) WEP (Wired Equivalent Privacy)
- b) WPA (Wi-Fi Protected Access)**
- c) SSL/TLS (Secure Sockets Layer/Transport Layer Security)
- d) AES (Advanced Encryption Standard)

4. What is the purpose of a VPN (Virtual Private Network) in a network security context?

- a) Encrypting network traffic to prevent eavesdropping**
- b) Connecting multiple LANs (Local Area Networks) over a wide area network (WAN)
- c) Authenticating users and controlling access to network resources
- d) Reducing latency and improving network performance

Section 2: True or false

5. Patch management is the process of regularly updating software and firmware to address security vulnerabilities and improve system performance.

ANS-**TRUE** Patch management means updating software to fix vulnerabilities and improve performance

6. A network administrator should perform regular backups of critical data to prevent data loss in the event of hardware failures, disasters, or security breaches.

ANS-**TRUE** regular backup protects important data from loss during failures or attacks.

7. Traceroute is a network diagnostic tool used to identify the route and measure the latency of data packets between a source and destination device.

ANS-**TRUE** Traceroute is used to see the path of packets and measure delay between source and destination.

Section 3: Short Answer

8. Describe the steps involved in conducting a network vulnerability Assignment.

ANS- A network vulnerability assessment is done to find security weakness and fix them before attackers can use them.

1. Make an inventory of all network devices

First, you make a list of everything connected to the network-computer, laptops, router, switch, printers, server etc. This helps you understand what exactly you need to protect.

2. Gather important technical information

For each device, collect details like

- Ip address
- Operating system version
- Software installed
- Services running
- Open ports (this gives a complete picture of the network setup.)

3. Run Vulnerability scanning tools

Tools such as Nmap, Nessus, OpenVAS, or even Windows Defender can scan the network. this tool automatically checks for:

- outdated software
- weak passwords
- open ports that should not be open
- missing security patches
- default admin credential

4. analysis the scan report

After scanning you get a list of vulnerabilities. You need to review the list to understand

- which problem are dangerous
- which problem are low risk
- which problem can be ignored

5. Rank vulnerabilities by improving importance

- we sort them by severity:
- high/critical → need immediate fix
- medium → fix soon
- low → fix, if possible, this help manage time and error better.

6. Apply fixed and correct action

- Based on the priority list you:
- Update software
- Install patches
- Change weak passwords
- Close unused ports
- Disable unnecessary service
- Modify firewall setting

7. Verify by rescanning the network

After fix issue you can run the scan again this ensures them inabilities are really gone and the network is now more secure

Section 4: Practical Application

9. Demonstrate how to troubleshoot network connectivity issues using the ping command.

ANS- The ping command is used to check if two devices can communicate, it sends small packets of data and waits for a reply.

Here's the detailed step-by-step process:

1. Check internet connectivity using a public IP

Ping 8.8.8.8

This tests if your pc can reach Google's DNS server.

- If you get replies → internet is working
- If not → problem with ISP or connection

2. Check domain name resolution

Ping Google.com

If you can ping IP but not a domain, it means:

- DNS server is not working
- DNS setting is incorrect

3. Check connection to router/gateway

Example:

Ping 192.168.1.1

This checks if your device can reach the router.

If this fails, the issue might be:

- Ethernet cable unplugged
- Router is down

4. Ping other devices on the same network

Example:

Ping 192.168.1.12

This checks if devices inside the LAN can communicate.

If it fails:

- Devices may be off
- Firewall blocking pings
- Wrong IP configuration

5. Understand ping results

Ping Result

Meaning

Reply from...	Connection successful.
Request timed out	No reply received blockage or offline devices.
Destination hosts unreachable	Devices not reachable or wrong IP.
Ping statistics (0%loss)	Network good.
Packet loss (30-100%)	Poor or unstable connection.

6. Physical checking if ping fails

If ping show errors, check:

- Ethernet cable connection
- Wi-Fi status
- Router Lights
- Network adapter status

Using ping helps you know exactly where the problem is- in the device, router, DNS, or the internet service.

Section 5: Essay

10. Discuss the importance of regular network maintenance and the key tasks involved in maintaining network infrastructure.

ANS- Regular network maintenance is necessary to keep a computer network secure, fast, and reliable. A network is used by many users daily for internet access, communication, data transfer, and sharing resources just like machine need regular servicing and a network also need proper care to avoid failure and security risks.

One major reason for maintenance is **security**. cyber threat such as hacking, viruses, malware and ransomware is becoming more common if a network is not protected attackers can steal data damage system or interrupt services. By performing regular updates and installing patches we fix security bug make the network harder to break into.

Another important reason is **performed**. over time network may slow down due to outdated configuration heavy traffic or hold hardware maintenance help keep the network running smoothly by removing bottleneck reducing latency and optimizing bandwidth usage. This ensures the user experience faster and smoothly connectivity.

Data protectionist another benefit on maintenance if a network crash or get hacked important data can be lost by tracking regular backup data can be restored quickly and the organization can continue its work without major disruption move down

Maintenance can also help in **preventing downtime** when small problems are ignored they can turn into big failure regular checkups help detect early warning signs such as failing cables or overheating device so they can fix before they cause damage. this keeps network available and dependable.

- **Security protection**
Hackers constantly look for network weaknesses. Regular maintenance helps fix security issues before someone attacks the system.
- **Prevents network failures**
Maintenance finds problems early- before they cause bigger damage.
- **Better performance**
It keeps data flowing smoothly, avoids slow speeds, and reduces delays.
- **Data safety**
Regular backup and checks prevent data loss due to failure or cyber-attacks.
- **Reduced costs**
Fixing small issues early is cheaper than repairing major failure later.

Main tasks involved in network maintenance

1. **Software and firmware updates**
Systems, antivirus, routers, and firewalls need to be updated with latest patches to close security holes.
2. **Monitoring the network**
Track network usage, check for unusual traffic, and watch for suspicious behaviour.
3. **backing up important data regularly**
backups ensure that even if a system crashes, the data can be restored.
4. **Checking hardware condition**
Make sure the devices are working well. Replace damaged cables or old equipment.
5. **User access control**
Allow only authorized users to access network resources.
Remove old, unused accounts.
6. **Network documentation**
Keep a record of:
 - Device list
 - IP addresses
 - Network layout
 - Login credentialsGood documentation helps during troubleshooting.

7. Testing disaster recovery plans

Practice how to restore system after failure so downtime is minimized.

8. Optimizing network configuration

Remove unsend services, close open ports, and network faster and safer.

Conclusion

Regular network maintenance is essential for any organization. Its provides security, reliability and performance. it makes sure that the network stay safe from the attacks runs efficiently and avoid costly downtime. Good maintenance both users and data protected and help the organization operate smoothly